

Index Number:

Candidate Name: _____

CT Group: 07S_____

HWA CHONG INSTITUTION



JC2 PRELIMINARY EXAMINATIONS

H2 BIOLOGY 9747

PAPER 1

11 Sept 2008

1 hr 15 min

INSTRUCTIONS TO CANDIDATES

Do not open this multiple choice booklet until you are told to do so.

Write your **name**, **CT** and **Index Number** on this cover page and the optical mark sheet in the spaces provided.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the **optical mark sheet**.

Read very carefully the instructions on the optical mark sheet.

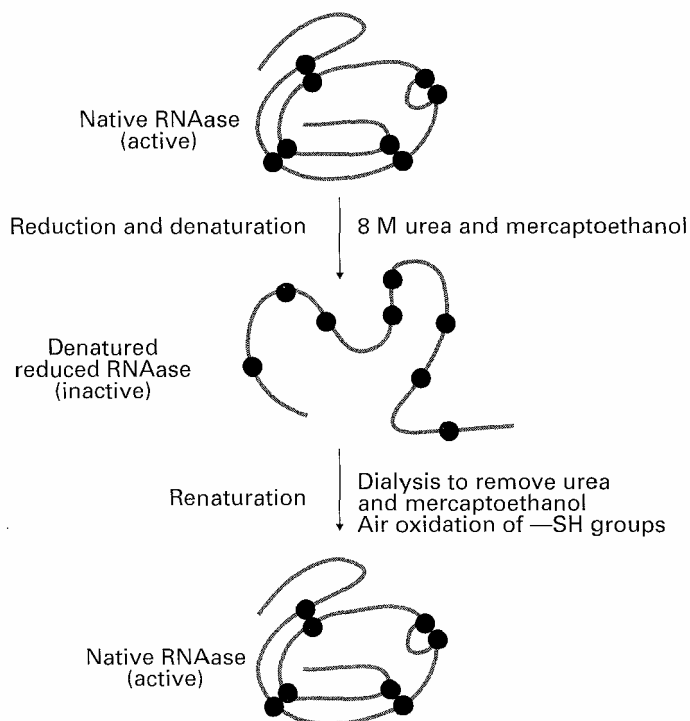
A NIL return is required. You are required to submit the optical mark sheet **only**.

INFORMATION FOR CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

- 1 An experiment was carried out to denature a pancreatic ribonuclease, RNAase as shown below. A reducing agent, mercaptoethanol, and urea were added. Denaturation was observed. Upon removal of urea and mercaptoethanol, renaturation occurred.



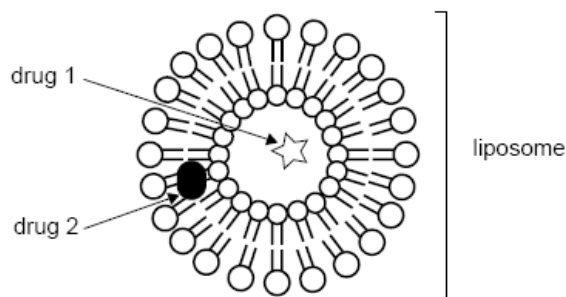
The following statements describe how polypeptides achieve their specific folding to give a tertiary structure based on the experiment.

I	The polypeptide chain must have refolded as directed by its primary sequence and that once refolded, the original disulphide bonds must have formed again.
II	Given the right conditions, denatured proteins can be refolded.
III	The tertiary structure of a protein can be predicted from its sequence.
IV	Denaturation alters the primary structure of a protein and destroys its biological activity.

Which of the statements are correct?

- A I and II
- B II and IV
- C II and III
- D I and IV

- 2 Liposomes can be found in the cytosol and are formed from a double layer of phospholipid molecules identical to those found in plasma membranes. Liposomes can also be manufactured and used to carry medicinal drugs into cells.

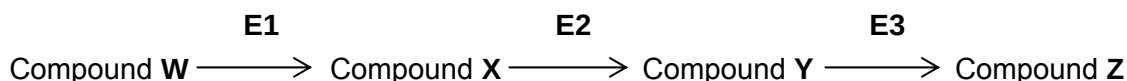


By examining the figure above, it is reasonable to expect that

- A drug 1 is lipophilic.
- B drug 2 is water soluble.
- C the interior of the liposome is aqueous.
- D liposome-based membranes have a rigid structure.

Questions 3 and 4 refer to the information below.

The flowchart below shows part of an enzyme-catalyzed sequence which involves 4 compounds (**W – Z**) and 3 enzymes (**E1 – E3**).



- 3 The addition of another compound **M** results in no change in the concentration of compound **W**, an accumulation of compound **X**, and a near absence of both compound **Y** and compound **Z**.

What does this information indicate about the mode of action of compound **M**, and which is a method by which the production of compound **Z** can be increased in the presence of compound **M**?

	Mode of action of compound M	Method to increase production of compound Z
A	It inhibits E1 .	Increase concentration of compound X in the reaction mixture.
B	It activates E1 .	Increase concentration of E2 in the reaction mixture.
C	It inhibits E2 .	Increase concentration of compound Y in the reaction mixture.
D	It inhibits E3 .	Increase concentration of compound Y in the reaction mixture.

- 4 The K_m value of **E3** for compound **Y** is 1.1. The K_m value of **E3** for compound **Y** in the presence of compound **N** is 1.5. Which of the following statements are correct?

I	Original V_{max} cannot be reached in the presence of compound N .
II	Compound N is a competitive inhibitor.
III	Binding of compound Y alters the globular 3D-structure of E3 .
IV	The final amount of product formed is the same in both the absence and presence of compound N .

- A I and III
 B II and IV
 C I, II and III
 D II, III and IV
- 5 In yeast, the secretion of an extracellular enzyme, invertase, involves two organelles **P** and **Q**. Three yeast secretory mutant strains were studied by monitoring for the presence of invertase outside of the yeast cells and also in **P** and **Q**.

The following table shows the observations made:

Yeast strain	Mutation(s)	Organelles in which invertase was detected	Invertase secreted out of yeast cell
1	Gene X only	Q only	No
2	Gene Y only	P and Q	No
3	Genes X and Y	Q only	No
Wild-type	None	P and Q	Yes

Which of the following deductions about the points of involvement of genes **X** and **Y** in the invertase secretory pathway could be correct?

- A
 Gene X
 Organelle P → Organelle Q → secreted outside cell
 Gene Y
- B
 Gene X
 Organelle Q → Organelle P → secreted outside cell
 Gene Y
- C
 Gene Y
 Organelle P → Organelle Q → secreted outside cell
 Gene X
- D
 Gene Y
 Organelle Q → Organelle P → secreted outside cell
 Gene X

6 The figure shows an electron micrograph of a cell.



Which of the following about structures **P** – **R** are correct?

	P	Q	R
A	Provides large surface area for attachment of ribosomes	Modification of transcripts	Active condensation of chromatin
B	Transport of proteins to Golgi apparatus	Formation of ATP from light energy	Active transcription of genes
C	Synthesis and processing of membrane proteins	Sequestering of protons within a specialised compartment	Active transcription of genes
D	Prevents proteins from degradation by cytoplasmic enzymes	Allows the oxidation of pyruvate	Active replication of DNA

- 7 The figure below shows the various ions, molecules and transport proteins that are present in a cholinergic synapse.

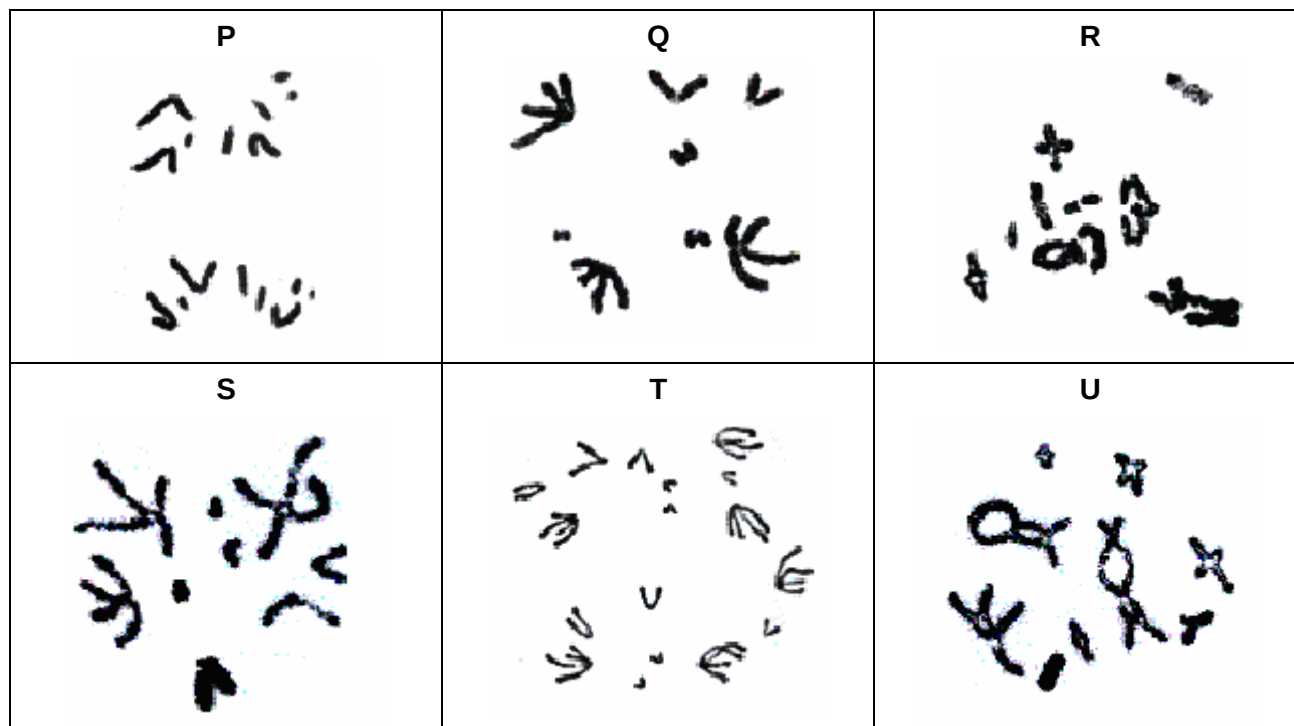


Which of the following statements correctly describe the transport of substances across synaptic membranes?

I	Some substances may diffuse readily across the membrane.
II	Ions move across the membrane through protein channels via facilitated diffusion.
III	Some ions are actively transported across the membrane by carrier proteins.
IV	There are at least two transport mechanisms that require energy expenditure.

- A II and III only
 B I, II and III only
 C I, III and IV only
 D All of the above

- 8 The diagram below shows some of the events that take place during meiosis.



Which of the following shows the correct sequence of events?

- A $U \rightarrow T \rightarrow R \rightarrow S \rightarrow Q \rightarrow P$
- B $U \rightarrow R \rightarrow T \rightarrow S \rightarrow Q \rightarrow P$
- C $R \rightarrow Q \rightarrow U \rightarrow S \rightarrow P \rightarrow T$
- D $R \rightarrow U \rightarrow Q \rightarrow S \rightarrow T \rightarrow P$
- 9 A researcher cultured skin cells in a flask. For these cells to survive *in vitro*, they have to adhere to the base of the flask. Dead cells are removed by throwing away the culture medium after three days. After a few days, the researcher observed that two layers of cells are formed on the base of the flask. If he were to carry out further experiments, he would **not** expect the cells to
- A have large nuclear to cytoplasmic ratio.
- B form well-defined tumour boundary.
- C be well-differentiated.
- D have regular nuclear shape.

- 10 In studying a virus, you find the following proportions of nitrogenous bases present: adenine 23%, guanine 37%, cytosine 23%, uracil 17%. Which of the following statement(s) regarding this virus is / are correct?

I	It probably uses RNA as its genetic material.
II	The genetic material of this virus is probably single stranded.
III	Base pairing rules in this virus include adenine : cytosine.

- A I only
 B I and II only
 C II and III only
 D All of the above

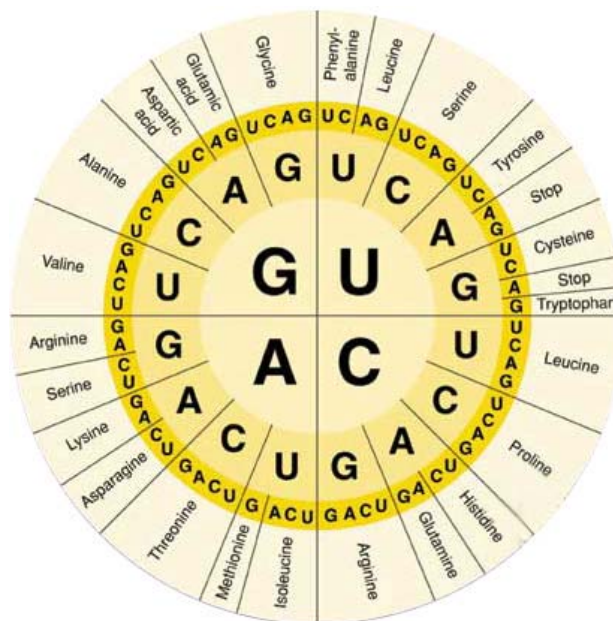
- 11 A pentapeptide has the following amino acid sequence:



where

Met : Methionine
 Leu : Leucine
 Trp : Tryptophan

Ala : Alanine
 Phe: Phenylalanine

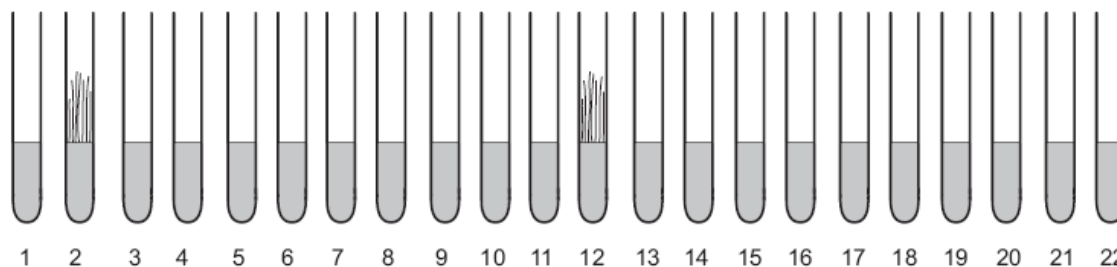


Using the genetic code shown in the diagram above, deduce the corresponding template strand sequence during transcription.

- A 5' – GAA GGC CCA CAA CAT – 3'
 B 5' – TTA GAA GGC CCA CAA CAT – 3'
 C 5' – ATG TTG TGG GCC TTT TAG – 3'
 D 5' – AUG CUG UGG GCA UUC – 3'



- 12** Fungi can make their own amino acids and are able to grow on a substance called minimal media. An abnormal strain of fungus that could not grow on minimal media was discovered. An experiment was designed and the results are outlined in the following diagram. The fungus grew in tube 12. The amino acid histidine had been added to this tube.



Tube 1 Minimal media

Tube 2 Minimal media plus all 20 different amino acids added

Tubes 3 to 22 Minimal media + one of the 20 amino acids. Each tube contained a different amino acid.

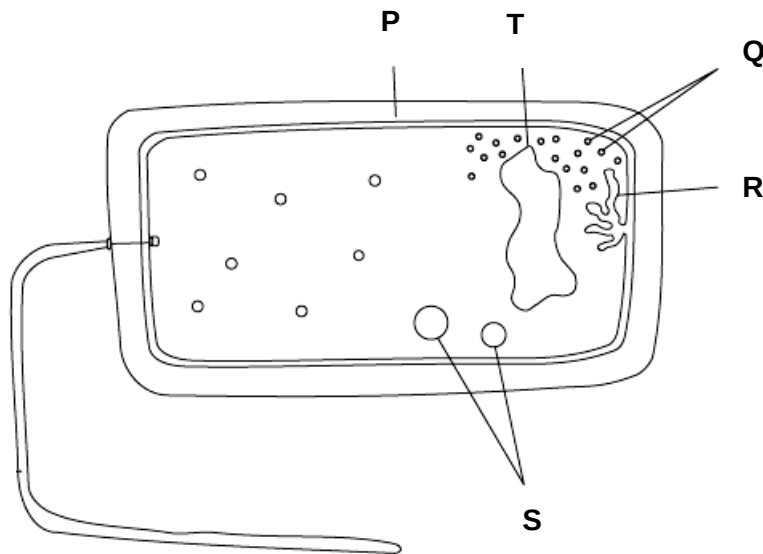
Spore of abnormal fungus was added to surface of media in each tube. All tubes were incubated under the same conditions for several days and then examined.

From this experiment, what conclusions can be made about the ability of the abnormal fungus to produce amino acids?

I	The abnormal fungus cannot produce histidine.
II	The abnormal fungus can produce the other 19 amino acid.
III	The control tubes present in this experiment were not necessary.

- A** I and II only
- B** II and III only
- C** I and III only
- D** All of the above
- 13** The genome of rhabdoviruses consists of a single-stranded RNA molecule whose sequence is complementary to the RNA sequence which functions as a messenger RNA. How is the “+” messenger RNA produced in such cells by rhabdovirus?
- A** Host cell RNA polymerase activity.
- B** One portion of the infecting RNA is directly translated by host cell ribosomes.
- C** Reverse transcriptase activity.
- D** The infecting virus particle contains an RNA-dependent RNA polymerase.

- 14 An *E. coli* cell, which carries a lambda prophage, is immune to a lytic infection by a second lambda virus because
- A proteins from the prophage's lytic genetic program prevent replication of the second virus.
 - B the second virus cannot inject its DNA.
 - C proteins from the prophage's lysogenic genetic program prevent replication of the second virus.
 - D the second virus cannot adsorb to the cell.
- 15 The figure below shows the general structure of a bacterial cell.



Use the letters on the diagram to identify the structures described below.

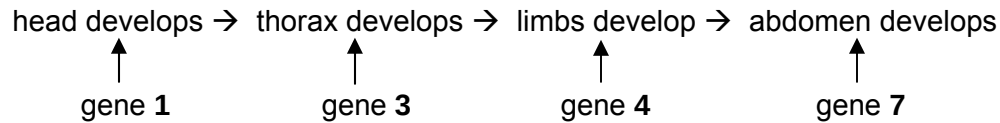
	Contains advantageous genes	Sites of protein synthesis	Prevents cell lysis	Contains enzymes for respiration
A	S	Q	P	R
B	S	T	R	Q
C	T	R	P	Q
D	T	S	P	R

- 16 X-chromosome inactivation
- A normally takes place in males but not females.
 - B is the cause of the Y chromosome being genetically inactive.
 - C takes place in humans so that the same X chromosome is inactive in all of the cells of a female.
 - D results in genetically turning off one of the two X chromosomes in female mammals.

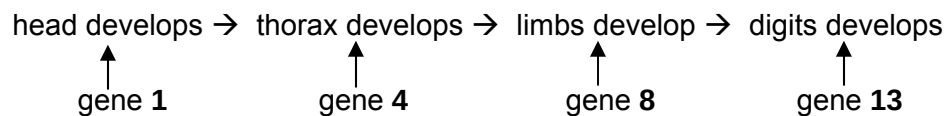


- 17** The normal order of development of features in an embryo is determined by gene sequences. The relevant groups of genes have been studied extensively in insects and vertebrates. These groups of genes are responsible for directing the sequential embryonic development of insects and vertebrates. In the summary below, gene activity starts at the left and continues in sequence to the right. The action of each gene in the group promotes the action of the subsequent gene.

In insects :



In vertebrates :



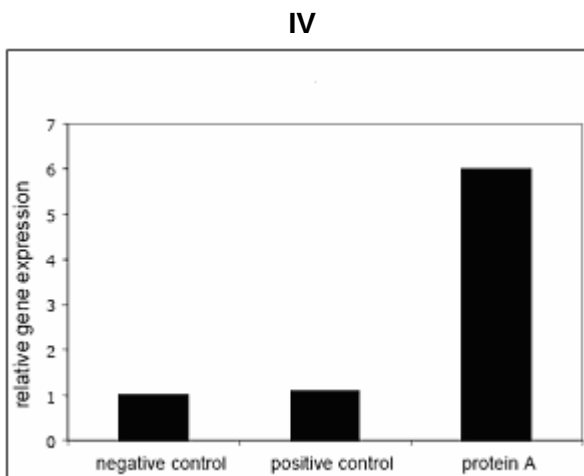
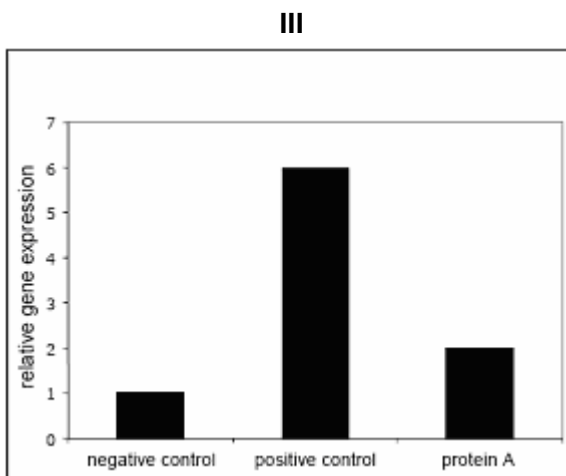
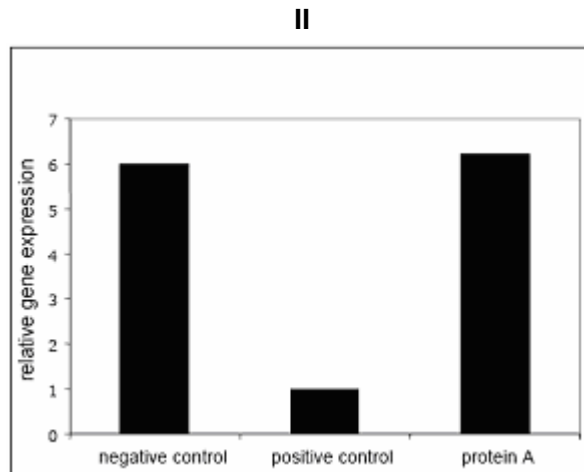
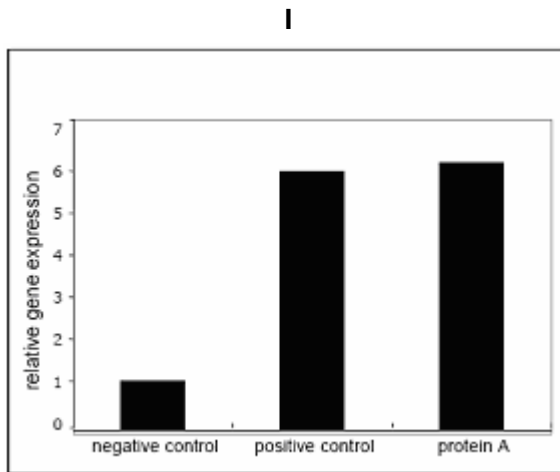
The above information suggests that if the action of a particular gene in a sequence was blocked

- A** an abnormal insect could develop possessing a head but no limbs.
 - B** an abnormal insect could develop possessing a thorax and limbs but no head.
 - C** an abnormal vertebrate could develop with a thorax but no head.
 - D** an abnormal vertebrate could develop possessing digits but no limbs.
- 18** About 12,000 genes are expressed in both chick liver and oviduct. However an estimated additional 5,000 genes are expressed only in liver, while an additional 3,000 genes are expressed only in oviduct. Which of the following could explain these observations?

I	The additional genes may have different methylation patterns in different tissues.
II	The concentrations of transcriptional enhancer elements for the additional genes vary in different tissues.
III	The number of genome copies is different in different somatic cells.
IV	A common set of genes are expressed for normal functions in liver and oviduct.

- A** I only
- B** II and III only
- C** I and IV only
- D** II and IV only

- 19 In an experiment testing the expression of a gene in a bacterial culture, it is found that the presence of protein A activates transcription of the gene. Which of the following graphs (I – IV) best represents the data that shows this result?



- A Graph I
B Graph II
C Graph III
D Graph IV

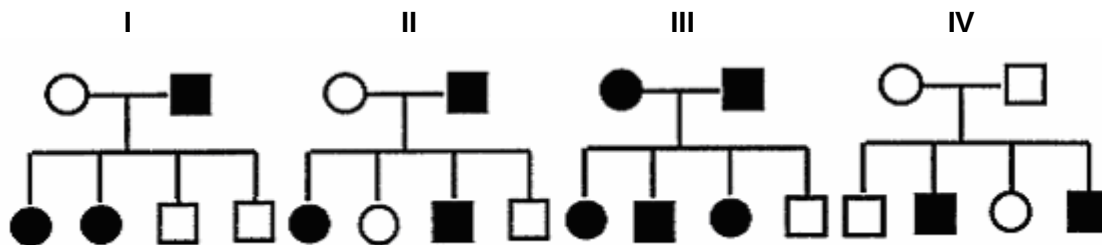


- 20 In lentils, the seed coat pattern is determined by a gene with 3 alleles. The phenotypes are marbled, spotted and clear. 4 crosses were repeated many times. The crosses and the outcomes of these crosses are shown in the table below.

Cross	Parents	Offspring phenotype and ratio
1	marbled x marbled	3 marbled : 1 clear
2	spotted x clear	1 spotted : 1 clear
3	marbled x marbled	3 marbled : 1 spotted
4	clear x clear	all clear

From the data, it is possible to conclude that

- A spotted is recessive to clear.
 - B all of the clear offspring are heterozygous.
 - C two thirds of the marbled offspring in cross 3 are heterozygous.
 - D the marbled parents in cross 1 have the same genotype as the marbled parents in cross 3.
- 21 The figure below shows 4 pedigrees. A shaded circle represents an affected female while a shaded square represents an affected male.

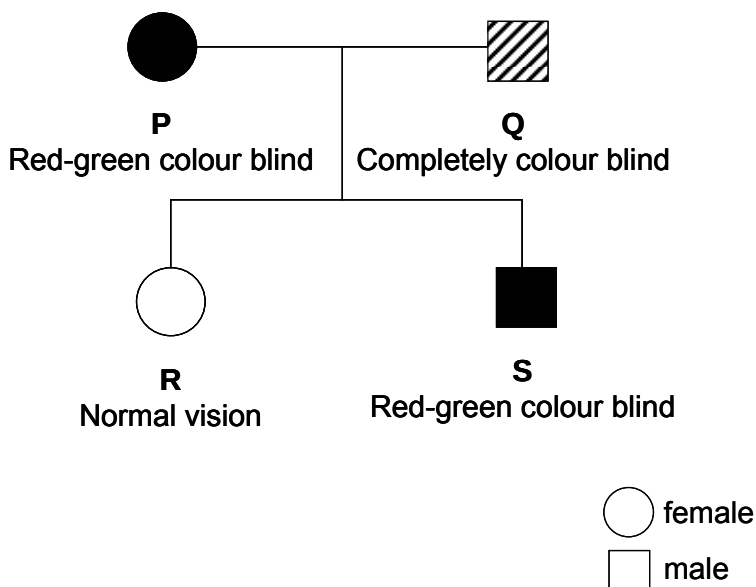


Which pedigree(s) show(s) a sex-linked dominant trait in humans?

- A I only
- B I and III
- C II and III
- D III and IV

- 22 Red-green colour blindness is controlled by a gene on the X chromosome. The allele for colour blindness, **g**, is recessive to the allele for normal colour vision, **G**. Complete colour blindness is controlled by a different gene which is not on the X chromosome. The allele for the development of normal cones (pigment cells in the retinal layer of the eye), **B**, is dominant to the allele for no cone development, **b**.

The figure below shows the phenotypes of members of a different family in which both types of colour blindness occur.



Which of the following are possible genotypes for individuals P, Q and S?

	P	Q	S
A	BBX^GX^G	bbX^GY	BBX^gY
B	BbX^GX^g	BbX^gY	BbX^GY
C	BBX^gX^g	bbX^GY	BbX^gY
D	BbX^GX^g	BbX^gY	BBX^GY

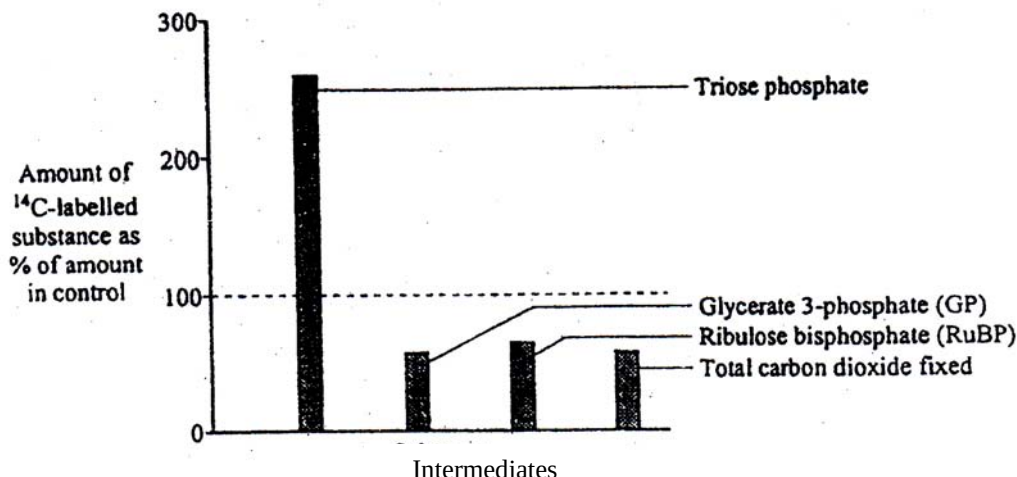
- 23 In the fruit fly, recessive mutations in either of 2 independently assorting genes, brown and purple, prevents the synthesis of red pigments in the eyes. Thus, homozygotes of either of these mutations have brownish-purple eyes. However, heterozygotes for both of these mutations have dark red, that is, wild-type eyes.

Crossing two flies which are both homozygous for a different mutation and heterozygous for the other corresponding mutation produced 144 offspring. How many offspring would be expected to be mutants?

- A 63
- B 72
- C 96
- D 108



- 24 An illuminated suspension of unicellular alga *Chlorella* in cadmium chloride was treated with carbon dioxide labelled with ^{14}C . The light was switched off and the amount of radioactivity present in some intermediates was determined after 10 minutes in the dark. A control suspension of cells without cadmium chloride was treated in exactly the same way. The bar chart below shows the amount of radioactivity in these intermediates in the cells from the cadmium chloride solution compared with the control cells.



From the above data, which of the following deductions can be made?

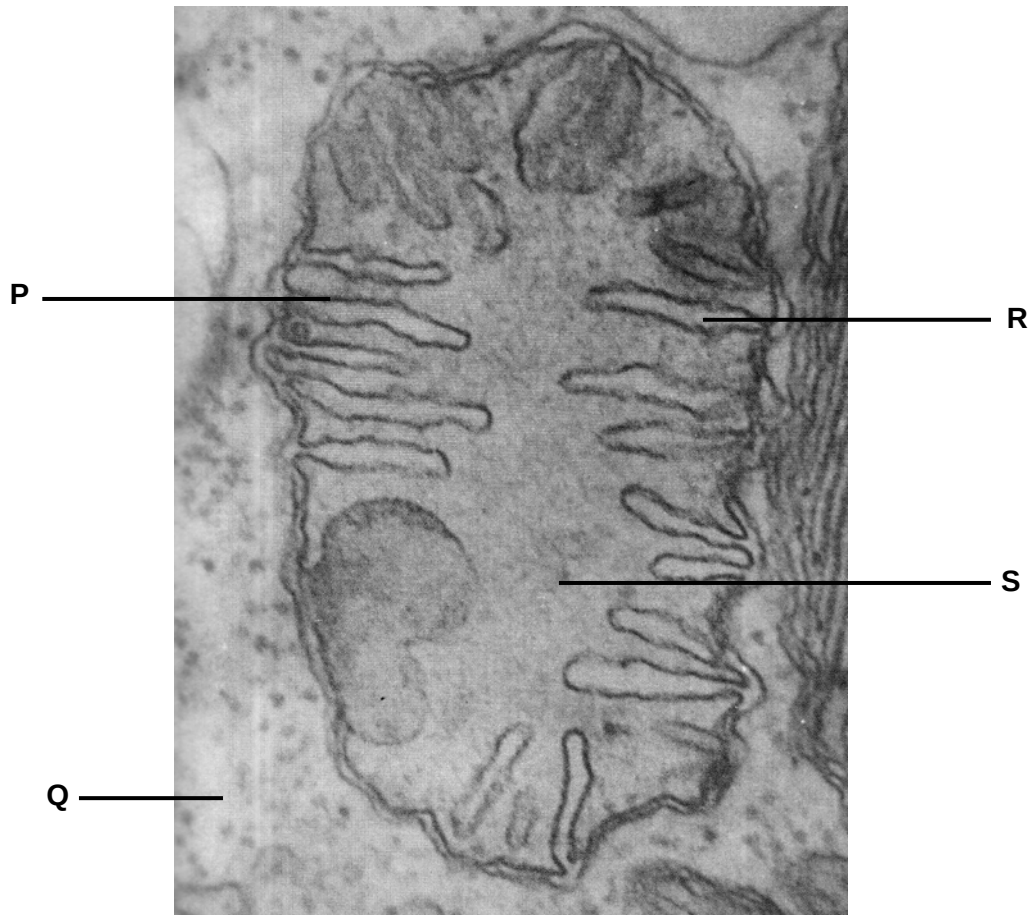
- A Cadmium chloride causes triose phosphate to accumulate as it prevents triose phosphate from being converted to hexose sugars.
 - B Cadmium chloride increases the rate at which glycerate-3-phosphate is being reduced to triose phosphate.
 - C Cadmium chloride prevents the regeneration of RuBP by inhibiting the enzyme at the stage just after triose phosphate.
 - D Cadmium chloride decreases the total amount of carbon dioxide fixed by RuBP by inhibiting RuBP carboxylase oxygenase.
- 25 In an experiment, chloroplast extracts were first treated with a chemical that 'snatches' away the electron that was accepted by the electron acceptor in photosystem I. The extracts were then treated with 2 hours of light and were provided with ample carbon dioxide and water.

Which of the following correctly shows the products that were formed after the experiment?

	O_2	ATP	Reduced NADP	Glucose
A	+	+	—	—
B	—	+	+	+
C	+	—	—	—
D	—	—	+	—

Key: (+) = present, (—) = absent

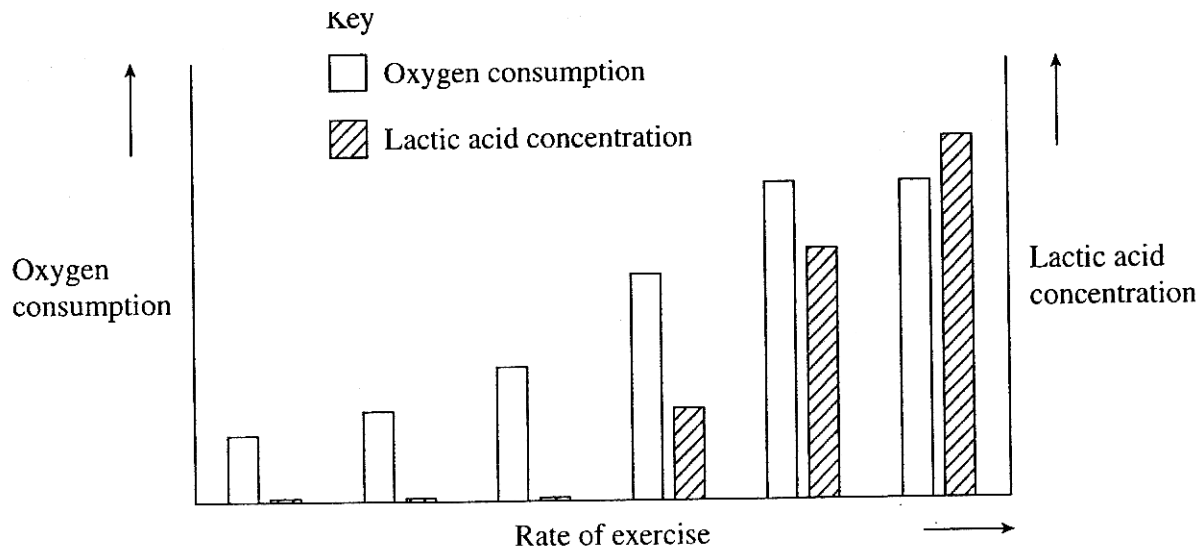
26 The figure below shows an electron micrograph of an organelle.



Match the following processes with the structures labelled P – S above.

	Breakdown of fructose-6-phosphate	Oxidative phosphorylation only	Temporary lowering of pH	Formation of water
A	Q	R	S	P
B	R	S	P	Q
C	S	P	Q	R
D	Q	P	R	S

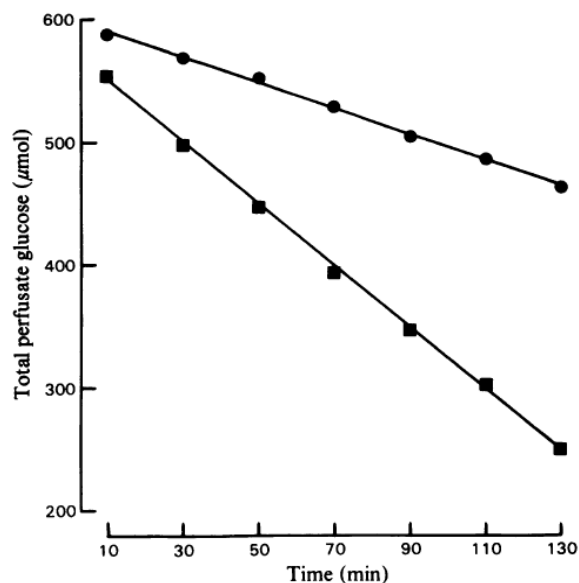
- 27 Scientists measured the oxygen consumption of an athlete at different rates of exercise. They also measured the lactic acid concentration in the athlete's muscles. The results are shown in the graph below.



Which of the following can be deduced from the information in the graph?

- A Anaerobic respiration occurs at a constant rate.
- B At a higher rate of exercise, muscle cells switch to anaerobic respiration.
- C Oxygen consumption increases proportionally with increased rate of exercise.
- D Increase in lactic acid concentration is due to increased oxygen consumption.

- 28 The figure below shows the time course of glucose uptake by the working rat heart in the absence of insulin (●) and in the presence of insulin (■). The lower the amount of total perfusate glucose, the greater is the uptake of glucose by the rat heart.

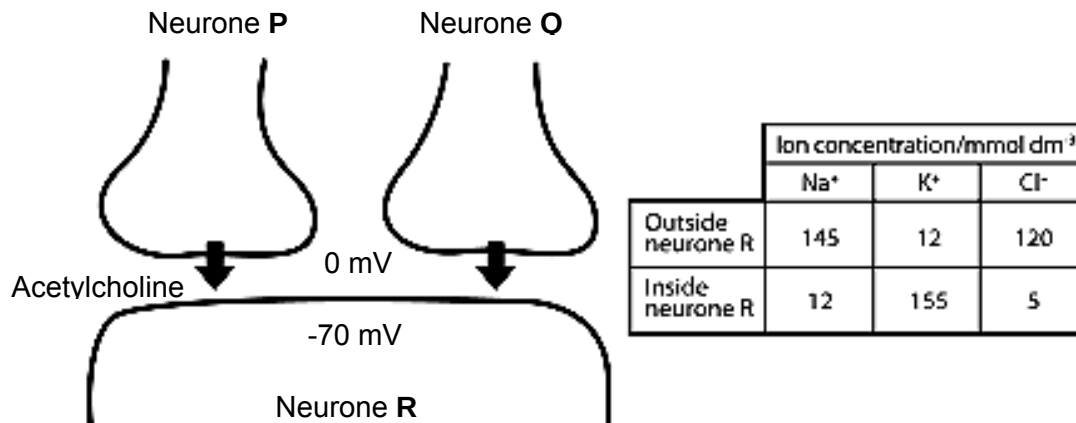


From the given data, which of the following statements can be concluded to be correct?

I	Cells in the rat's heart have receptors for insulin.
II	There is increased uptake of glucose by the rat's heart in the presence of insulin.
III	Release of insulin is stimulated by increase in blood glucose level.
IV	The rat is given a carbohydrate-rich meal prior to the experiment.

- A I and II only
 B I and IV only
 C II and III only
 D III and IV only

- 29 The figure below shows axon terminals from each of two neurones, **P** and **Q**, forming synapses with a third neurone, **R**. When stimulated, neurone **P** releases acetylcholine while neurone **Q** releases a structurally different neurotransmitter, N2. Concentrations of some ions are also shown.



Which of the following statements best explains why an action potential is not triggered in neurone **R** when both neurones **P** and **Q** are stimulated?

- A N2 competes with acetylcholine for receptor binding at the postsynaptic membrane of neurone **R**, resulting in less Na⁺ ions entering to cause depolarization.
 - B N2 causes the membrane potential of neurone **R** to be less negative, therefore more acetylcholine must be released by neurone **P** to cause greater influx of Na⁺ ions.
 - C N2 increases the permeability of the postsynaptic membrane of neurone **R** to Cl⁻ ions, resulting in hyperpolarization.
 - D Acetylcholine is released slowly from neurone **P** while N2 is released rapidly from neurone **Q**.
- 30 Three proteins isolated from a human cell were investigated for their involvement in cell signalling pathways.

	Protein A	Protein B	Protein C
ATPase Activity	+	-	-
GTPase Activity	-	-	+
DNA binding	-	+	-
Transmembrane domain	+	-	-

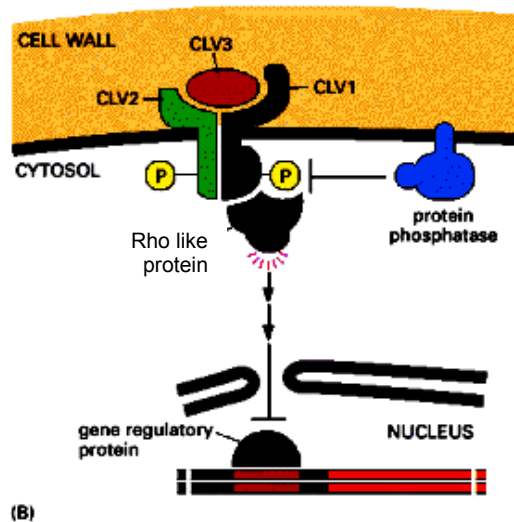
Key: (+) = present, (-) = absent

Which of the following correctly shows the identity of these three proteins?

	Protein A	Protein B	Protein C
A	Na ⁺ -K ⁺ Pump	Ras protein	Testosterone Receptor
B	Testosterone Receptor	Ras protein	Na ⁺ -K ⁺ Pump
C	Na ⁺ -K ⁺ Pump	Testosterone Receptor	Ras protein
D	Ras Protein	Testosterone Receptor	Na ⁺ -K ⁺ Pump



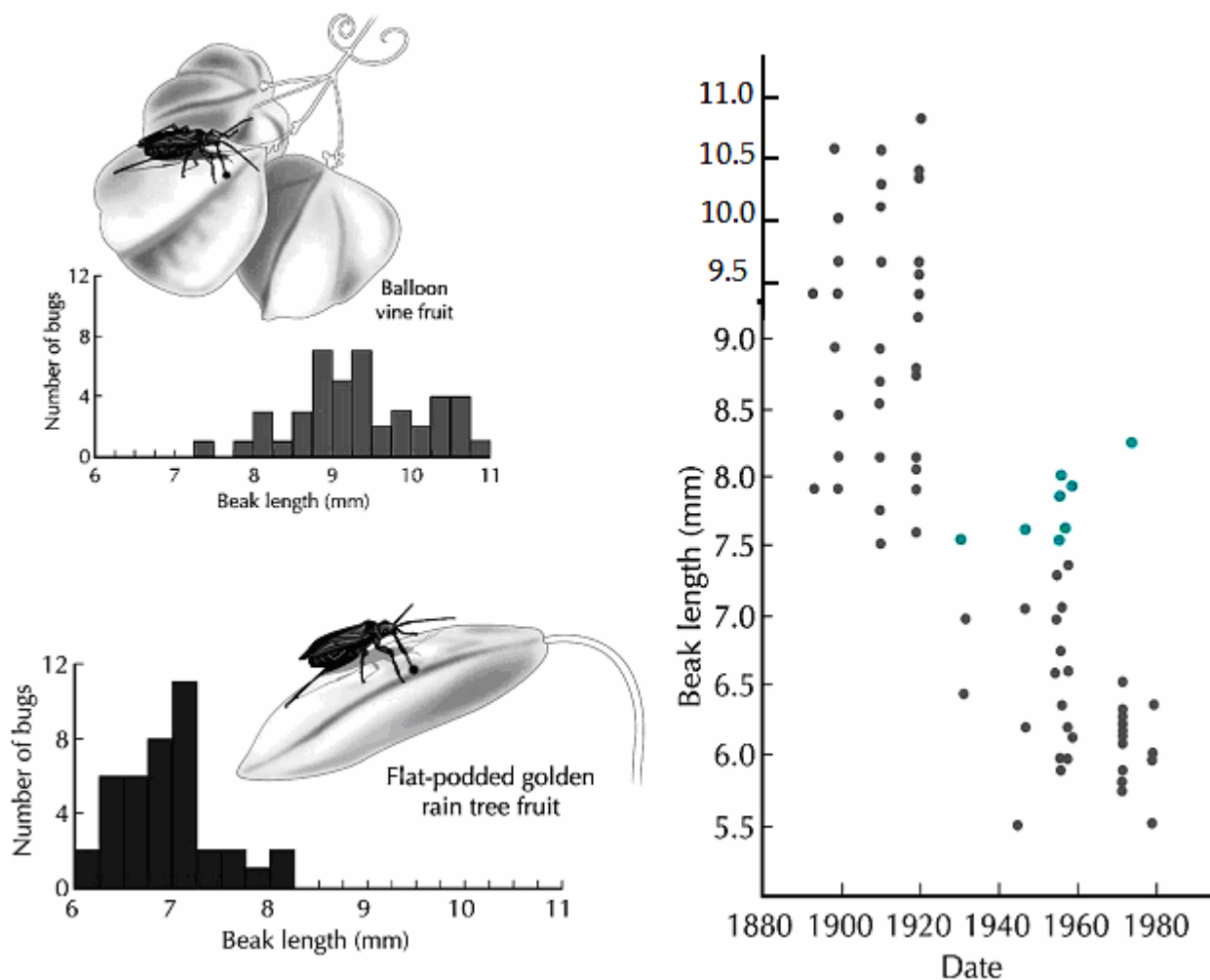
- 31 The diagram below shows the cell signalling pathway involved in cell proliferation and differentiation in the shoot meristem.



Which of the following sequences regarding the activation of the above signalling pathway is correct?

- A Binding of CLV3 to subunits CLV2 and CLV1 → dimerisation of CLV2 and CLV1 → autophosphorylation of serine/threonine residues → activation of Rho-like protein → activation of gene transcription.
 - B Dimerisation of CLV2 and CLV1 → binding of CLV3 to dimer → autophosphorylation of serine/threonine residues → activation of Rho-like protein → activation of gene transcription.
 - C Binding of CLV3 to subunits CLV2 and CLV1 → dimerisation of CLV2 and CLV1 → dephosphorylation of serine/threonine residues by protein phosphatase → activation of Rho-like protein → activation of gene transcription.
 - D Dimerisation of CLV2 and CLV1 → binding of CLV3 to dimer → dephosphorylation of serine/threonine residues by protein phosphatase → activation of Rho-like protein → activation of gene transcription.
- 32 In the summer of 1995, at least 15 iguanas survived Hurricane Marilyn on a raft of uprooted trees. They rode the high seas for a month before colonizing the Caribbean island, Anguilla. These few individuals were perhaps the first of their species, *Iguana iguana*, to reach the island. When one such healthy iguana was mated with another from its source of origin in the Virgin Islands, the offspring were infertile. Which of the following best explains this?
- A Founder effect
 - B Artificial selection
 - C Allopatric speciation
 - D Diversifying selection

- 33 Soapberry bugs in America originally fed on the native balloon vine using their sharp beaks to penetrate the fruit. In the 1920s, the flat-podded golden rain tree was introduced from Asia. This has thinner-skinned fruit.



What can you conclude from the data presented above?

I	Soapberry bugs with shorter beaks are being selected for.
II	Soapberry bugs evolved shorter beaks after they switched to feed on this new host plant.
III	Soapberry bugs had a preference for the introduced flat-podded golden rain tree.
IV	Soapberry bugs population contains the allele for short beaks since 1900.

- A I only
 B I and II
 C I, II and III
 D II, III and IV

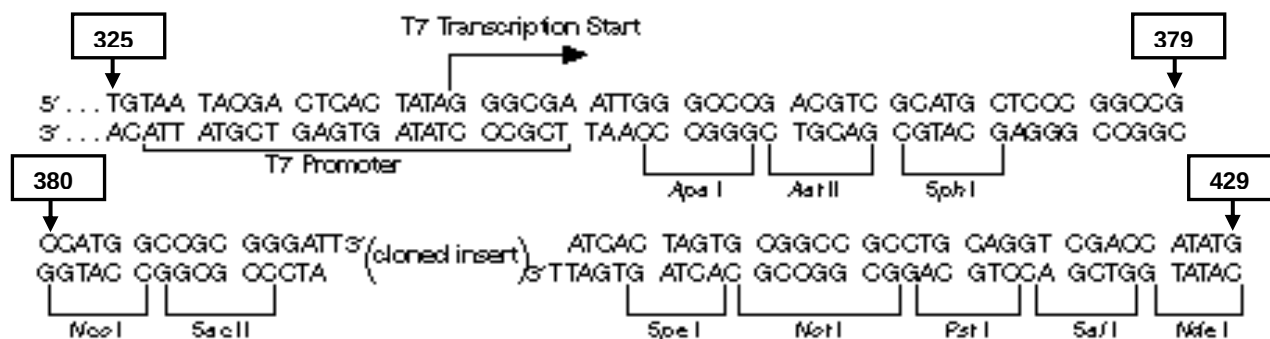
34 Which of the following ideas were **not** part of Darwin's theory of evolution by natural selection?

I	Acquired variation is inherited.
II	Variation between individuals arises by gene mutation.
III	Organisms produce more offspring than the environment can support.
IV	Only those individuals best adapted to the environment survive and reproduce.
V	Regions that encode portions of the polypeptide that are vital for structure and function are less likely to incur mutations.

- A I, II and V
 B I, III and IV
 C II, III and V
 D II, IV and V

Questions 35 and 36 refer to the information below.

pGEM-T is a 7.25kb plasmid cloning vector often used in creating recombinant DNA. It contains two selectable marker genes, *Tet^R* and *lacZ*. The figure below shows a portion of the multiple cloning site (MCS) of the cloning vector. It is located within the *lacZ* gene.



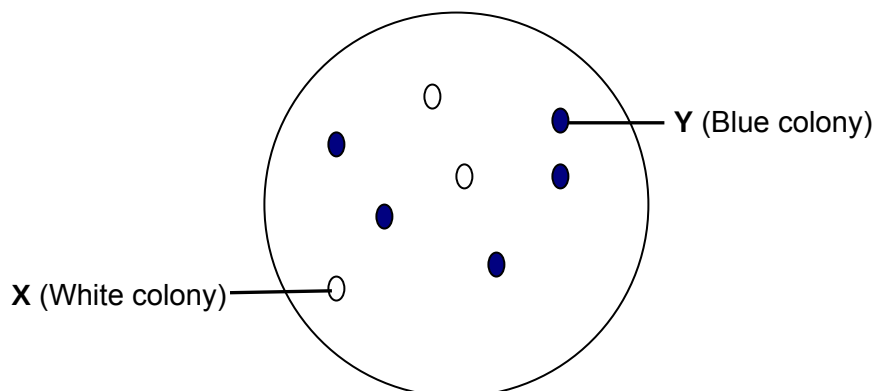
Note: The number in each box represents the nucleotide position within the plasmid.

35 Which of the following is the resulting number of fragments when pGEM-T is digested by the respective enzymes?

	SacII only	SacII and SpeI
A	2	3
B	2	2
C	1	3
D	1	2



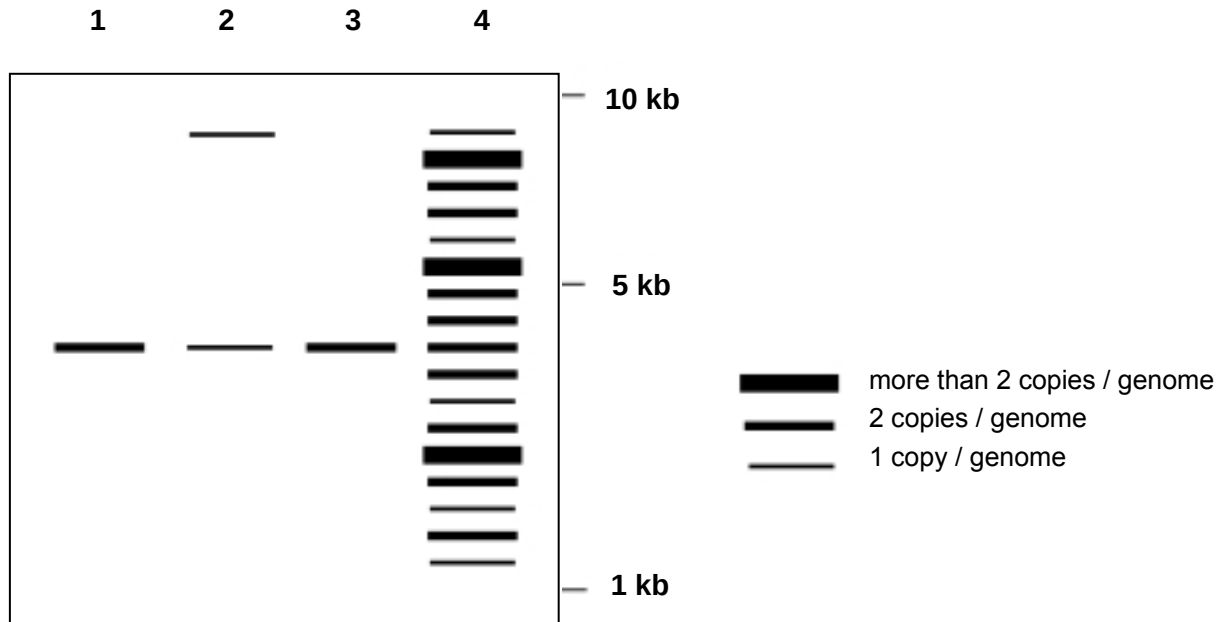
- 36 Using *Sac*II as the restriction enzyme, a 2.31kb insert (gene of interest) is cloned into the MCS as shown in the figure on Page 21 and the recombinant plasmid is transformed into *E. coli* cells. The transformants are then plated on medium containing tetracycline and X-Gal. The following result is obtained.



Colonies X and Y are isolated and the plasmid DNA from the colonies is extracted. The extracted plasmids are then digested with *Sac*II. Which of the following correctly shows the fragments obtained?

	Colony X	Colony Y
A	A 9.56kb fragment	A 7.25kb fragment
B	A 7.25kb fragment	A 9.56kb fragment
C	7.21kb and 0.34kb fragments	7.21kb and 2.31kb fragments
D	7.21kb and 0.34kb fragments	7.25kb and 0.34kb fragments

- 37 The *ras* oncogene is involved in a variety of human cancers. DNA isolated from a number of normal and cancerous human tissues were digested with *Eco*RI. A Southern blot was performed and the membrane was probed with radioactive-labelled cloned *ras* DNA. After exposure to x-ray film, the resulting autoradiograph is shown below.



Lane	Sample
1	white blood cells from a healthy human
2	lymphoma cells (cancerous)
3	bladder carcinoma cells (cancerous)
4	sarcoma cells (cancerous)

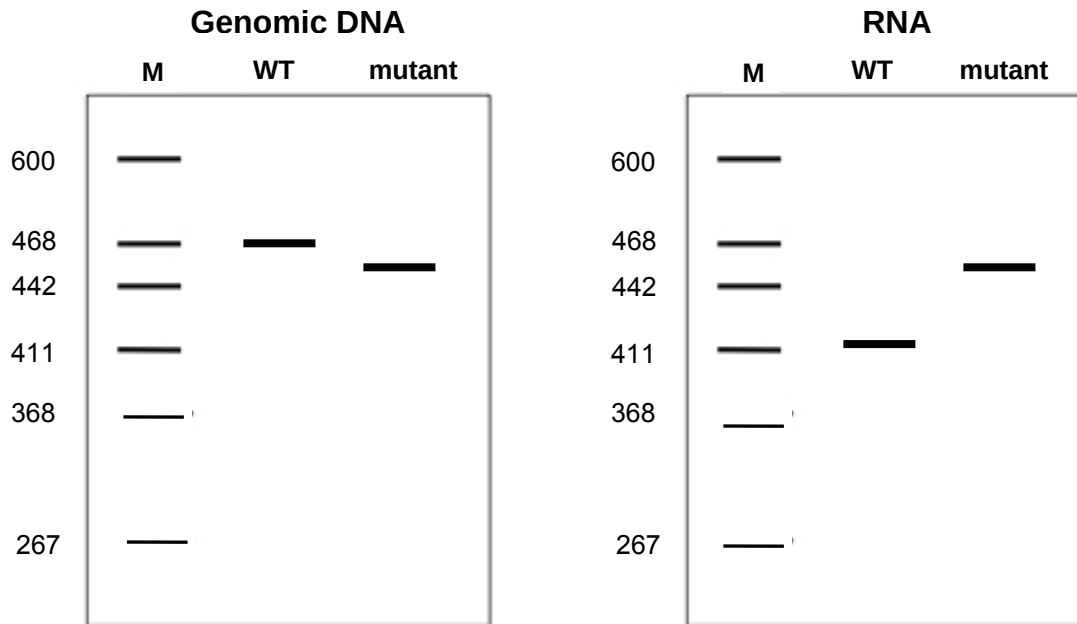
Based on the experimental results, which of the following statement(s) could be deduced?

I	In tissue 3, there was no mutation within the <i>ras</i> coding region.
II	The result for tissue 2 could be explained by separate point mutations that caused the removal of an <i>Eco</i> RI site and the creation of another within the probe DNA.
III	Gene amplification was involved in tissue 4.
IV	In tissue 3, the mutation did not affect existing <i>Eco</i> RI sites within the <i>ras</i> gene.

- A II only
 B I, III and IV
 C II, III and IV
 D III and IV

- 38** In separate reactions, PCR was performed using a pair of primers with sequences complementary to parts of separate exons of a gene suspected to be involved in a genetic disease.

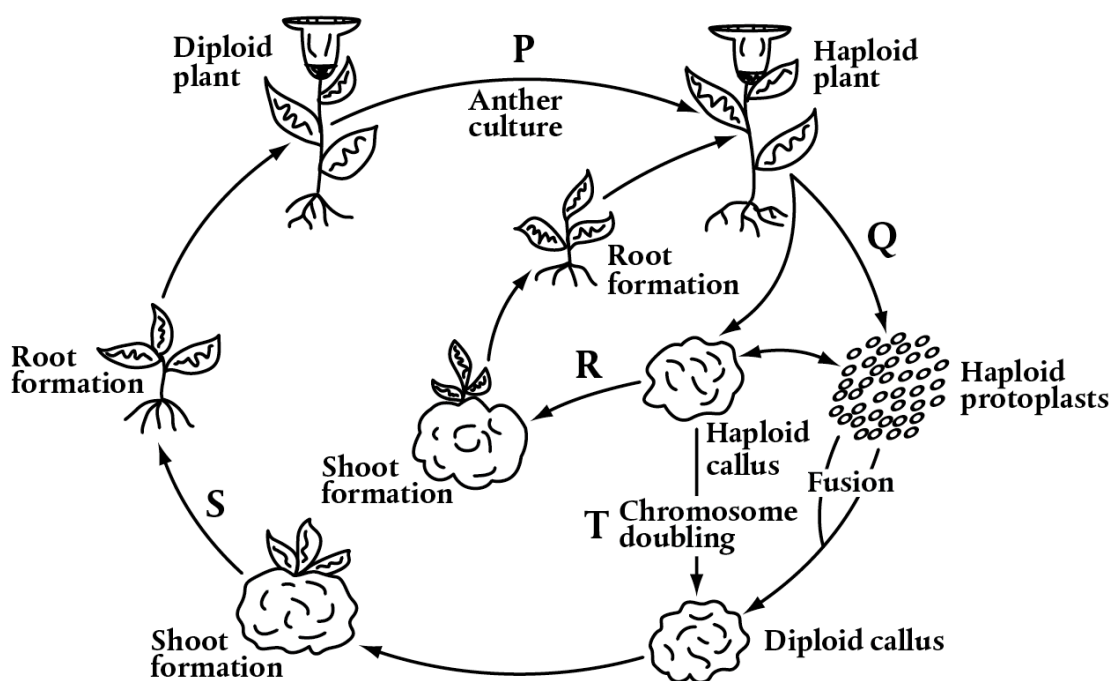
Using genomic DNA and RNA isolated from healthy (wild-type) and diseased cells, separate PCR reactions were done. The PCR products were analysed on polyacrylamide gels. The sizes of the bands in the molecular weight marker (M) are indicated in base-pairs.



Which of the following best explains the results obtained?

- A** Deletion of several introns
- B** Deletion of an exon
- C** Deletion of a splice site
- D** Deletion of several codons

- 39 The figure below gives some of the possible experimental manipulations in the culture of cells from *Petunia* plants.



Which of the following correctly identifies the stage for each of the events taking place in the culture of *Petunia* plant cells?

	Meiotic division	Removal of cell wall	Restoration of somatic number of chromosomes	Removal and subsequent culture of individual cells from microshoots
A	R	P	S	Q
B	P	T	T	R
C	R	R	S	P
D	P	Q	T	S

- 40 Breeders have developed a variety of chicken that has no feathers. Which methods were most likely used to produce this variety?
- A Artificial selection and inbreeding
 - B Grafting and hybridization
 - C Regeneration and incubation
 - D Vegetative propagation and binary fission

--- End of Paper ---

